Member 3 - MRI Image Enhancement using CLAHE

IT25104064 | Hettiarachchi H.A.A.S

Assigned preprocessing part	MRI Image Enhancement using CLAHE
Colab reference	Combined Colab - MEMBER 3 section only

Simple presentation speech

Good morning. My preprocessing technique is MRI image enhancement using CLAHE.

CLAHE means Contrast Limited Adaptive Histogram Equalization. Its purpose is to improve local contrast, especially in regions where the difference between dark and bright areas is not very clear.

In our code, I first convert the RGB image into the LAB color space. I apply CLAHE only to the lightness channel, called the L channel, and then combine it back with the other channels. Our CLAHE settings use a clip limit of 2.0 and an 8 by 8 tile grid.

For EDA, I show the same MRI image before CLAHE and after CLAHE. I also show an intensity histogram before and after enhancement. The image comparison helps us visually see how local contrast changes, while the histogram shows how pixel intensities are redistributed.

After defining the method, I apply the same fixed CLAHE settings to the resized Training, Validation, and Testing images. Because this is a deterministic enhancement, we can apply it consistently to all three sets.

The important point is that CLAHE changes the image representation by improving local contrast. We should not automatically say it improves model accuracy. Its usefulness must be checked later using Validation results during model training.

Key result to remember: Before: resized MRI with its original local contrast. After: CLAHE changes local contrast and the intensity distribution using fixed settings.

Live Colab demonstration - before / run / after

Do not read the code line by line. Say the purpose first, run or show the cell, then explain the output.

Run / show	Before running - say	After output - say
Define CLAHE	I use CLAHE with clip limit 2.0 and an 8 by 8 tile grid on the LAB lightness channel.	Only brightness information is enhanced, and the image is converted back to RGB.
Image BEFORE CLAHE	First I show the resized MRI before contrast enhancement.	This is the baseline image used for the visual comparison.
Image + histogram AFTER CLAHE	Now I apply CLAHE and compare the image and intensity histogram.	The local contrast and pixel-intensity distribution change after CLAHE.
Apply to all splits	Finally I use the same fixed CLAHE settings for Training, Validation, and Testing.	All three splits receive consistent deterministic enhancement; there is no random transformation here.

What this member should clearly demonstrate

• Explain what the selected preprocessing technique means.

• Explain why it is needed for this Brain Tumor MRI dataset.

• Run or show the implementation code and its important output.

• Show at least one before/after EDA visualization and interpret what changed.